

Shlok Patel

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Boston, MA — Available Jan – Sep 2027

EDUCATION

Northeastern University, Boston, MA May 2029
Candidate for B.S. in Computer Science and Business Administration (Financial Technology) **GPA: 3.74/4.00**
Coursework: Software Engineering, Object-Oriented Design, Algorithms & Data Structures, Databases, Systems
Honors & Societies: Dean's List, Honors College, NU ACES, NU Disrupt

TECHNICAL SKILLS

Languages: Python, Java, SQL, C++, JavaScript, TypeScript, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, Flask, FastAPI, PyTorch, Pandas, NumPy, SciPy, scikit-learn
Developer Tools & Databases: MongoDB, PostgreSQL, AWS, Supabase, Prisma, Docker, Git, CI/CD

PROJECTS

PhishAware | *MongoDB, FastAPI, Python, TypeScript, Groq/Gemini* Jul 2026 – Sep 2026

- Led a 3-person team building a full-stack phishing-awareness platform across **6** attack vectors (**1000+** tests).
- Shipped **47 REST endpoints** with frontend types auto-generated from one **OpenAPI** spec via **Orval**.
- Built an **OIDC authentication** flow with **PKCE**, letting organizations sign in with their own identity provider.
- Trained a scikit-learn risk model (**$R^2 = 0.94$**) on **FastAPI** with a **Groq/Gemini** LLM pipeline and failover.

Rainfall Learning | *React, TypeScript, Yjs, Docker* Dec 2025 – Jul 2026

- Spearheaded UI on a real-time collaborative code editor, cutting edit conflicts to **2%** via **Yjs** CRDT sync.
- Optimized the UI layer with **10+** engineers, boosting render performance **20%** during concurrent editing.
- Containerized build/CI with **Docker**, cutting environment failures by **15%** with automated testing.

Derivative Hedging via Reinforcement Learning | *Python, PyTorch, Stable-Baselines3, NumPy* Jan 2026 – Mar 2026

- Engineered a reproducible harness (**6** strategies, **4** regimes, **10K**-resample paired bootstrap) for hedging policies.
- Proved the RL-vs-delta edge insignificant ($p=0.29-0.97$); **Whalley-Wilmott** beat it with **80%** fewer trades.
- Redesigned the reward as a **CVaR** objective, then traced via drift-ablation how it was gamed by a directional tilt.

EXPERIENCE

NU Intelligent Automation and IoT Research Group Jul 2026 – Present
Undergraduate Research Assistant advised by Prof. Sarita Singh Boston, MA

- Engineer an ML pipeline (**librosa**, **Praat**, **PyTorch**) fusing voice, tapping, and gait for Parkinson's detection.
- Architect gated and attention fusion nets plus a stacking meta-learner, benchmarking **8+** models leakage-free.
- Raise **AUROC** from **0.720 to 0.774** ($p=0.080$) across **2,685** subjects, hardened by **100+** unit tests.

Khoury College of Computer Sciences May 2026 – Present
Teaching Assistant, Program Design and Implementation II Boston, MA

- Lead **30+** weekly debugging sessions in **Java**, coaching students on OOP, data structures, and control flow.
- Teach **100+** students AI-assisted debugging and prompt engineering with **Claude** and **GitHub Copilot**.

Student Government Association Jan 2026 – Present
Software Engineer Boston, MA

- Develop a CMS letting non-technical editors publish to **5,000+** students, cutting publish time by **10%**.
- Architect version history, rollback, role-based access, and soft-delete archiving on a **Next.js/Prisma** stack.
- Engineer **Prisma**-validated API endpoints enforcing schema integrity on every mutation over **PostgreSQL**.
- Collaborate with a cross-functional team of **5+** to elicit stakeholder requirements and ship in **Agile** sprints.

INTERESTS & LEADERSHIP

Interests: Volunteering, Chess, Graphic Design, Walking, Observing Skylines
Leadership: Project-Series Mentor, NU Oasis — lead groups of 5+ on their first-ever software development journeys.